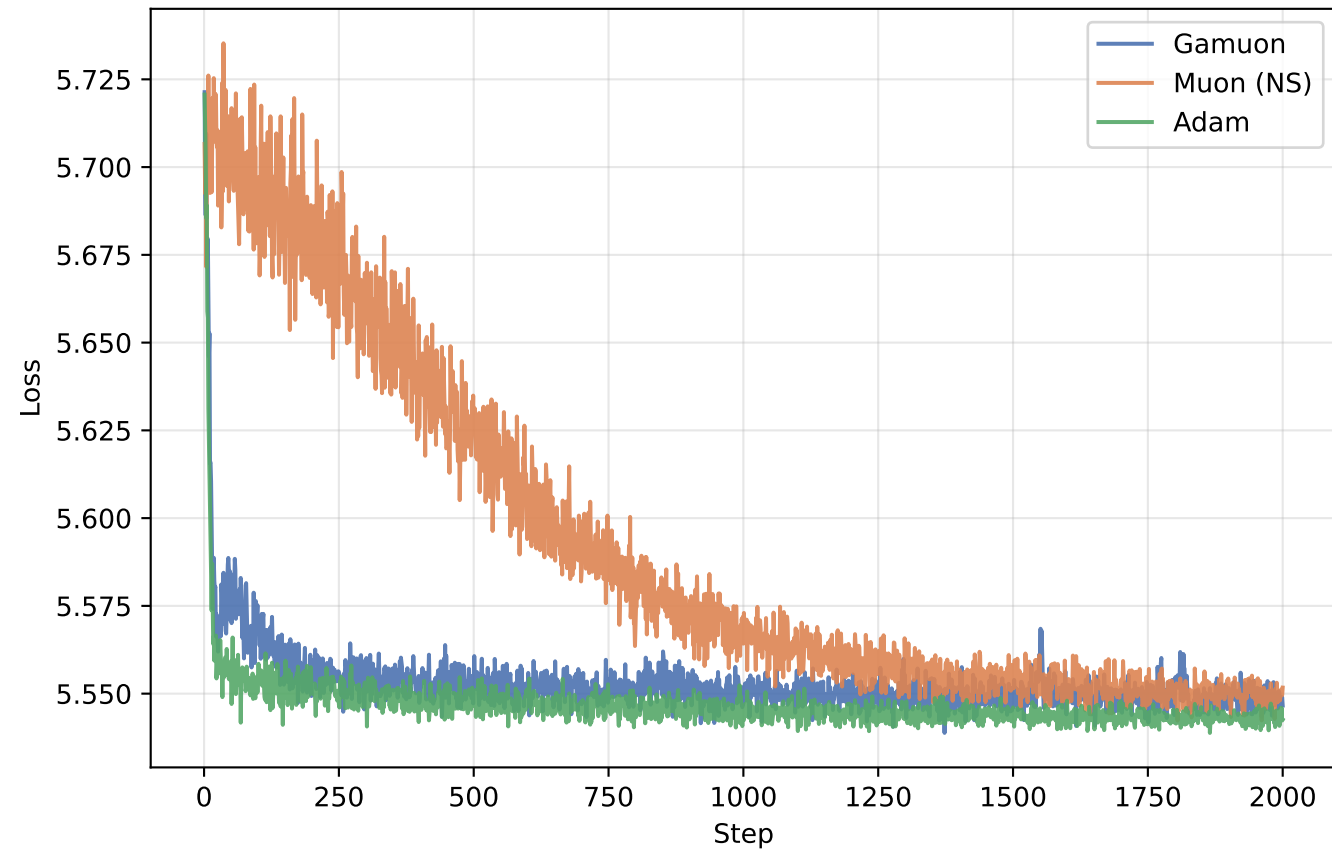
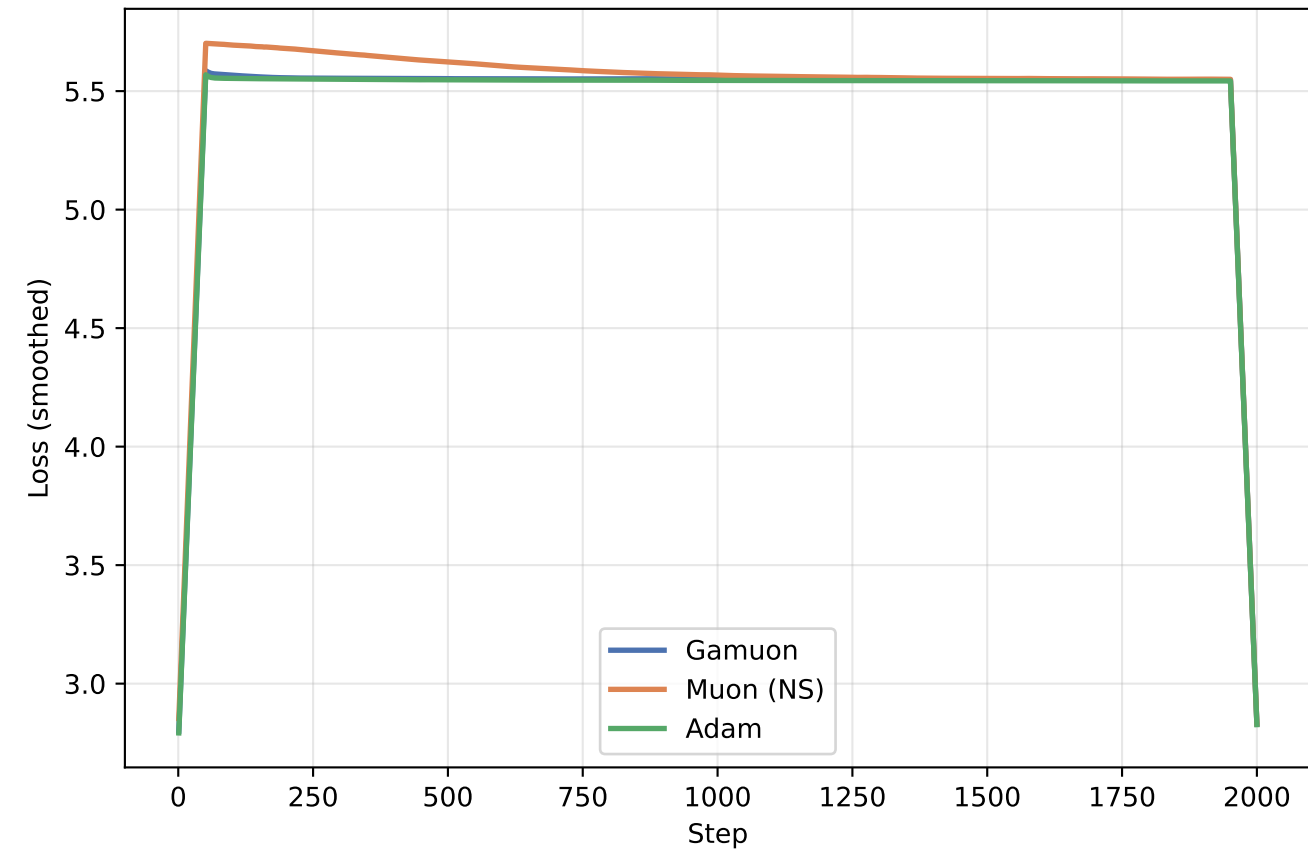


Gamuon vs Muon (Newton-Schulz) vs Adam — Small Transformer

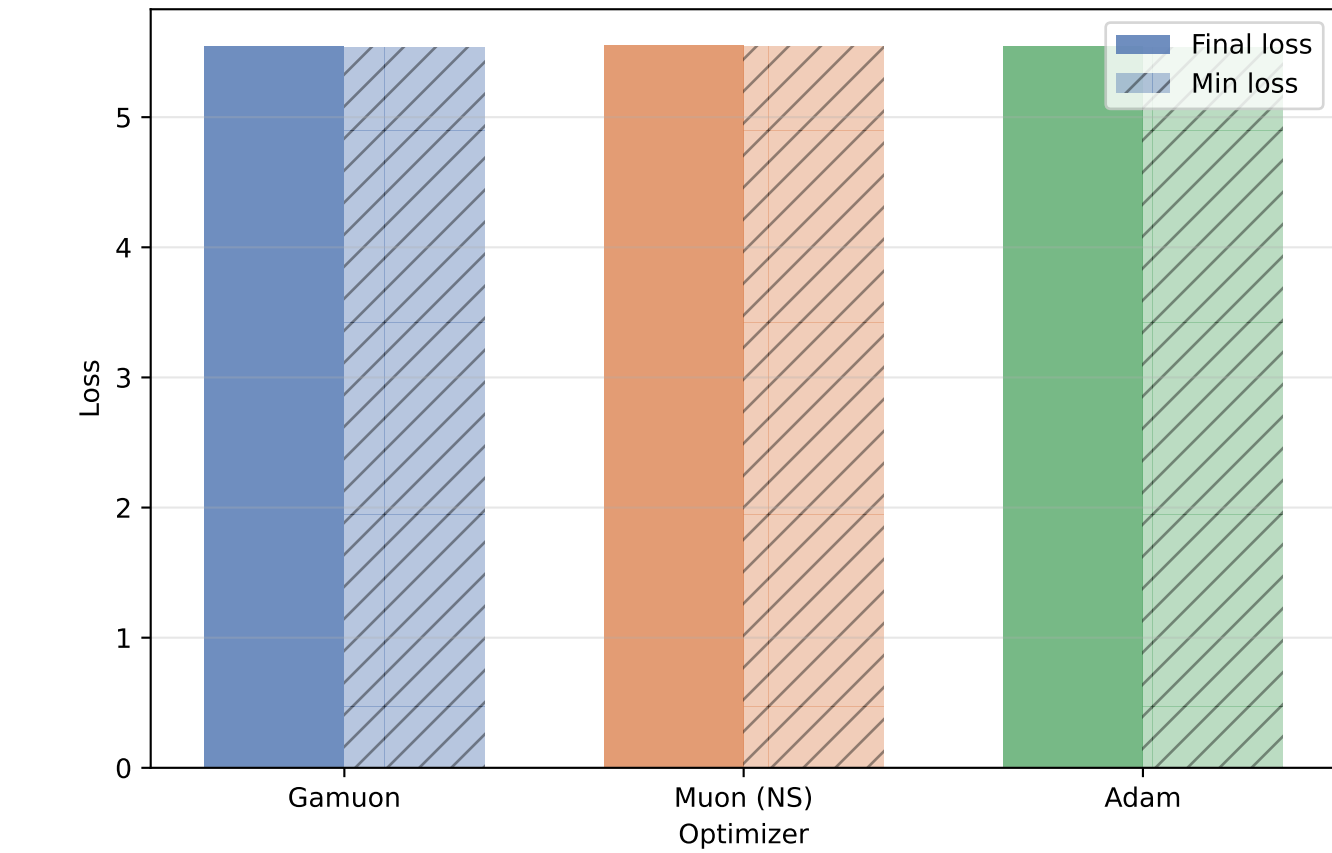
Training Loss vs Steps



Smoothed Loss (window=100)



Final & Minimum Loss



Per-Step Wall-Clock Time

